

A Input: a CSV of cross-validated predictions

environment	genotype	observed	predicted	method
BRK_00	PI182561	66.0	70.4	ridge_pc5_lo
BRK_00	PI382161	92.5	69.9	ridge_pc5_lo
BRK_00	PI382167	72.0	71.1	ridge_pc5_lo
...	...	...	...	...

33,116 rows in the example file

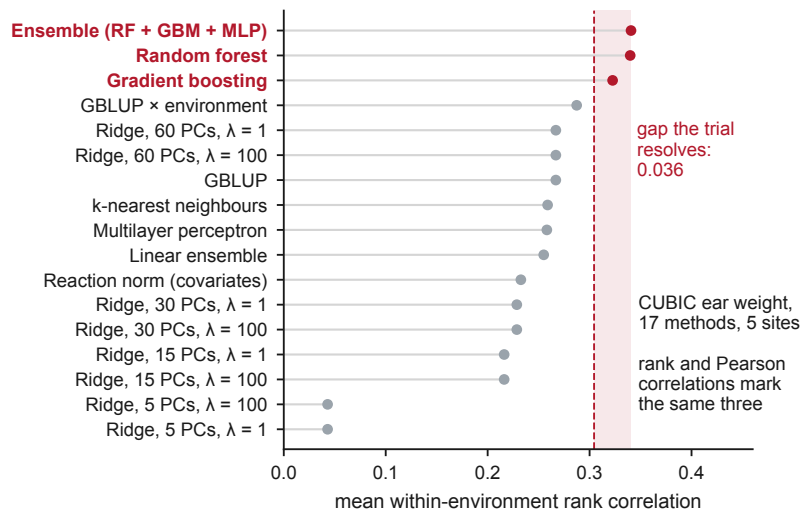
GPverdict
browser or Python;
data stay local

Verdict for the bundled spring wheat example

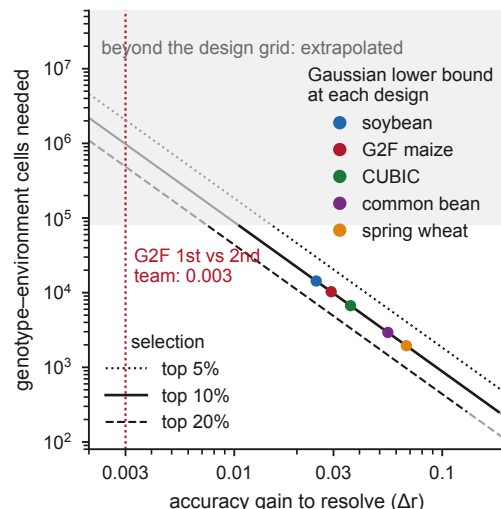
**17 methods, 109 environments, 384 genotypes;
top 10% selected within each environment**

- Both within-environment correlations (rank correlation and Pearson r) can rank the methods; RMSE and pooled metrics cannot.
- ridge_pc40_lo leads by the rank correlation, mlp by Pearson r. With 1,948 cells per method the smallest gap the trial resolves is 0.067, so it cannot separate ridge_pc40_hi, ens and gbm from them.
- An RMSE ranking would pick ens; RMSE and Pearson r order 25% of the 136 method pairs oppositely.
- Resolving a gain of 0.03 would need at least 9,773 cells (Gaussian lower bound).

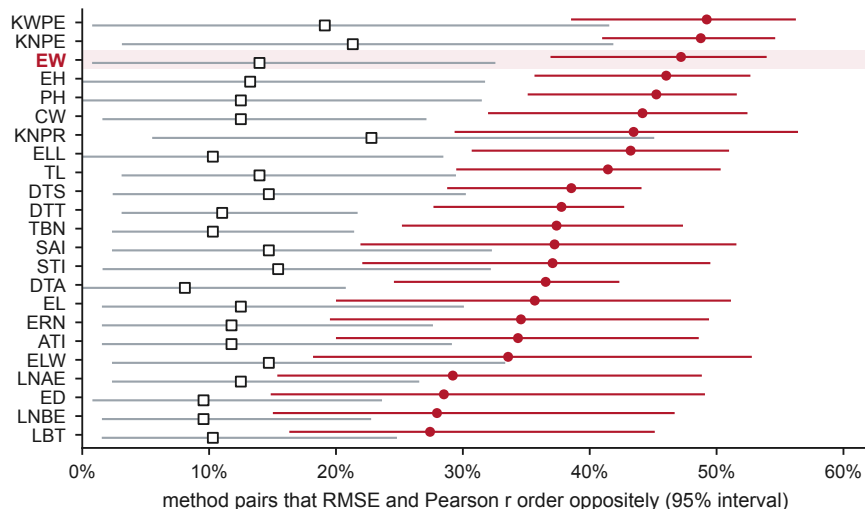
B



C



D



CUBIC: all 23 traits of
Supplementary Table S19,
same 52-method panel,
rule fixed before the run

ear weight (EW, panel B)
highlighted

- all 52 methods (with miscalibration variants)
- 17 base methods